TD 3

Part 2 - Verify Git identity

```
@Henri127 →/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $ git --version
git version 2.56.0
@Henri127 →/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $ git config user.name
Henri127
@Henri127 →/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $ git config user.email
henri.portier@edu.devinci.fr
```

Part 3 - Inspect the real TD02 changes

```
@Henri127 →/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $ git status
On branch Henri
Your branch is up to date with 'origin/Henri'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        labs/TD04_BRANCHES_AND_MERGE.md

nothing added to commit but untracked files present (use "git add" to track)
@Henri127 →/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $ git diff
@Henri127 →/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $ git diff
diff --git a/src/main.py b/src/main.py
index 19361dc..3d1316f 100644
--- a/src/main.py
+++ b/src/main.py
@@ -98,3 +98,4 @@ def main():

:...skipping...
diff --git a/src/main.py b/src/main.py
index 19361dc..3d1316f 100644
--- a/src/main.py
+++ b/src/main.py
@@ -98,3 +98,4 @@ def main():
```

Questions

Which files are modified?
Which lines were added?
Which lines were removed?
Does the diff match the work you intended?
Is there temporary debugging output that should be removed before committing?

Part 4 - Understand git status and git diff

```
@Henri127 →/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $ git status
git status --short
git diff
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   src/main.py

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        labs/TD04_BRANCHES_AND_MERGE.md

no changes added to commit (use "git add" and/or "git commit -a")
 M src/main.py
?? labs/TD04_BRANCHES_AND_MERGE.md
diff --git a/src/main.py b/src/main.py
index 19361dc..3d1316f 100644
--- a/src/main.py
+++ b/src/main.py
@@ -98,3 +98,4 @@ def main():

 if __name__ == "__main__":
     main()
+
\ No newline at end of file
```

Part 5 - Stage the intended change

```
@Henri127 →/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $ git add src/main.py
@Henri127 →/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $ git status
git diff --staged
On branch Henri
Your branch is up to date with 'origin/Henri'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   src/main.py

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        labs/TD04_BRANCHES_AND_MERGE.md

diff --git a/src/main.py b/src/main.py
index 19361dc..3d1316f 100644
--- a/src/main.py
+++ b/src/main.py
@@ -98,3 +98,4 @@ def main():

 if __name__ == "__main__":
     main()
+
\ No newline at end of file
```

Part 6 - Create the first meaningful commit

```
@Henri127 ➜/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $ git commit -m "Formatted main.py with a space"
[Henri 7c6783f] Formatted main.py with a space
 1 file changed, 1 insertion(+)
```

Part 7 - Read the local history

```
@Henri127 ➜/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $ git status
git log --oneline
On branch Henri
Your branch is ahead of 'origin/Henri' by 1 commit.
  (use "git push" to publish your local commits)

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        labs/TD04_BRANCHES_AND_MERGE.md

nothing added to commit but untracked files present (use "git add" to track)
7c6783f (HEAD -> Henri) Formatted main.py with a space
3560e3d (origin/Henri) TD 2
24f6b30 Merge pull request #1 from Henri127/main
c8e7706 Update TD Group and MarketPulse Team details
4cb5427 Modify team members in TEAM_TEMPLATE.md
5207db8 first td session
ba5c9f4 Update Readme
9fd6c8a Update Readme
bfb94a6 init projecttemplate
```

```
@Henri127 ➜/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $ git show HEAD
commit 7c6783f6234a9612a84d66b47295b9618d241b83 (HEAD -> Henri)
Author: Henri127 <henri.portier@edu.devinci.fr>
Date:   Fri Nov 13 15:15:38 2026 +0000

    Formatted main.py with a space

diff --git a/src/main.py b/src/main.py
index 19361dc..3d1316f 100644
--- a/src/main.py
+++ b/src/main.py
@@ -98,3 +98,4 @@ def main():

 if __name__ == "__main__":
     main()
+
\ No newline at end of file
```

Be able to explain:

which files changed;
what the diff represents;
why the commit message is appropriate;
who is recorded as the author.

Part 8 - Commit quality and traceability
Each student should eventually be able to answer:

Which commits are mine?

What did I change?
Why did I change it?
Can I explain the diff?
Can I reproduce the change?

Part 10 - CORE validation

```
@Henri127 →/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $ git status
git log --oneline
python src/main.py
On branch Henri
Your branch is ahead of 'origin/Henri' by 1 commit.
  (use "git push" to publish your local commits)

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        labs/TD04_BRANCHES_AND_MERGE.md

nothing added to commit but untracked files present (use "git add" to track)
7c6783f (HEAD -> Henri) Formatted main.py with a space
3560e3d (origin/Henri) TD 2
24f6b30 Merge pull request #1 from Henri127/main
c8e7706 Update TD Group and MarketPulse Team details
4cb5427 Modify team members in TEAM_TEMPLATE.md
5207db8 first td session
ba5c9f4 Update Readme
9fd6c8a Update Readme
bfb94a6 init projecttemplate
=== MarketPulse ===

Market configuration
Period: 1 month
Interval: Daily

Instruments
AAPL - Apple Inc.
```

```
=== MarketPulse ===

Market configuration
Period: 1 month
Interval: Daily

Instruments
AAPL - Apple Inc.
Observations : 21
First Close : 250.0 USD
Last Close : 266.2 USD

Benchmark
SP500 - S&P 500
Observations : 21
First Close : 6600.0
Last Close : 6742.0

2026-09-01
2026-09-30

Max instrument : 266.2
Min instrument : 249.8

Max benchmark : 6742.0
Min benchmark : 6598.0
```

```
@Henri127 →/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $ git show HEAD
commit 7c6783f6234a9612a84d66b47295b9618d241b83 (HEAD -> Henri)
Author: Henri127 <henri.portier@edu.devinci.fr>
Date:   Fri Nov 13 15:15:38 2026 +0000

    Formatted main.py with a space

diff --git a/src/main.py b/src/main.py
index 19361dc..3d1316f 100644
--- a/src/main.py
+++ b/src/main.py
@@ -98,3 +98,4 @@ def main():

 if __name__ == "__main__":
     main()
+
\ No newline at end of file
```

```
commit 3560e3dbb67eff80f900ae2482ee5b4bfa60c343 (HEAD -> Henri, origin/Henri)
Author: Henri127 <henri.portier@edu.devinci.fr>
Date:   Wed Oct 7 18:09:43 2026 +0000

    TD 2

diff --git a/src/main.py b/src/main.py
index 884b045..19361dc 100644
--- a/src/main.py
+++ b/src/main.py
@@ -25,6 +25,42 @@ def load_prices():
 def filter_prices(prices, ticker):
     return [row for row in prices if row["ticker"] == ticker]

+def get_first_close(prices):
+    return float(prices[0]["close"])
:
```

Optional 1 - Tracked vs untracked

```
@Henri127 →/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $ touch td03_notes.txt
git status
On branch Henri
Your branch is ahead of 'origin/Henri' by 1 commit.
  (use "git push" to publish your local commits)

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        labs/TD04_BRANCHES_AND_MERGE.md
        td03_notes.txt

nothing added to commit but untracked files present (use "git add" to track)
```

```
@Henri127 →/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $ rm td03_notes.txt
git status
On branch Henri
Your branch is ahead of 'origin/Henri' by 1 commit.
  (use "git push" to publish your local commits)

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        labs/TD04_BRANCHES_AND_MERGE.md

nothing added to commit but untracked files present (use "git add" to track)
```

Optional 2 - Stage one file while another remains modified

```
@Henri127 ➜/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $ git status
git diff
On branch Henri
Your branch is ahead of 'origin/Henri' by 1 commit.
  (use "git push" to publish your local commits)

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        labs/TD04_BRANCHES_AND_MERGE.md

nothing added to commit but untracked files present (use "git add" to track)
```

```
@Henri127 ➜/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $ git add /workspaces/esilv-marketpulse-starter-g07-tSHA/labs/TD04_BRANCHES_AND_MERGE.md
@Henri127 ➜/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $ git status
git diff
git diff --staged
On branch Henri
Your branch is ahead of 'origin/Henri' by 1 commit.
  (use "git push" to publish your local commits)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   labs/TD04_BRANCHES_AND_MERGE.md

diff --git a/labs/TD04_BRANCHES_AND_MERGE.md b/labs/TD04_BRANCHES_AND_MERGE.md
new file mode 100644
index 0000000..09be143
--- /dev/null
+++ b/labs/TD04_BRANCHES_AND_MERGE.md
@@ -0,0 +1,812 @@
+# TD04 - Branches + Merge
+
+## Duration
+
+```text
+1 hour 30 minutes
```

+

Business requirement

+

```text
+"Features should no longer be developed directly on main."
```

```
+"Features should no longer be developed directly on main."
+```
+
+## Context
+
+TD03 introduced local Git history.
+
+TD04 adds feature isolation.
+
+The new question is:
+
+> How can we develop a change without modifying the current main line until the change is ready?
+
+Git branches provide that isolation.
:
```

Optional 3 - Filter file history

```
@Henri127 ➜/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $ git log --oneline -- src/main.py
7c6783f (HEAD -> Henri) Formatted main.py with a space
3560e3d (origin/Henri) TD 2
bfb94a6 init projecttemplate
@Henri127 ➜/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $
```

Question:

What does this command filter?

Optional 4 - Graph view

```
@Henri127 ➜/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $ git log --oneline --graph --decorate
* 7c6783f (HEAD -> Henri) Formatted main.py with a space
* 3560e3d (origin/Henri) TD 2
*   24f6b30 Merge pull request #1 from Henri127/main
|\
| * c8e7706 Update TD Group and MarketPulse Team details
| * 4cb5427 Modify team members in TEAM_TEMPLATE.md
|/
* 5207db8 first td session
* ba5c9f4 Update Readme
* 9fd6c8a Update Readme
* bfb94a6 init projecttemplate
```

Optional 5 - Author filtering

```
@Henri127 ➜/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $ git log --author="Henri127" --oneline
7c6783f (HEAD -> Henri) Formatted main.py with a space
3560e3d (origin/Henri) TD 2
24f6b30 Merge pull request #1 from Henri127/main
c8e7706 Update TD Group and MarketPulse Team details
4cb5427 Modify team members in TEAM_TEMPLATE.md
@Henri127 ➜/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $
```

Optional 6 - Tracked files

```
@Henri127 ➜/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $ git ls-files
.gitignore
CONTRIBUTING.md
README.md
TEAM_TEMPLATE.md
config/settings.yml
data/sample/bloomberg_reference_expected.json
data/sample/bloomberg_reference_sample.json
data/sample/instruments.json
data/sample/prices.csv
evidence/README.md
labs/README.md
labs/TD01_BOOTSTRAP_LINUX.md
labs/TD02_PYTHON_CSV_JSON.md
labs/TD03_GIT_LOCAL_WORKFLOW.md
labs/TD04_BRANCHES_AND_MERGE.md
readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md
requirements.txt
src/main.py
@Henri127 ➜/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $
```